

# INSY6212 - Assignment 2

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## Group Members

| Name             | Student Number |
|------------------|----------------|
| Dylan Olsen      | ST10440133     |
| Tyreece Sewbaran | ST10439074     |
| Ndumiso Jwara    | ST10440124     |
| Ethan Edwards    | ST10442420     |
| David Forster    | ST10247775     |

# Question 1

## Question 1.1

### Importance of a Project Evaluation Form

An organized method for evaluating and rating possible projects prior to their official selection and start is a project evaluation form. It is crucial because it gives decision-makers an unbiased, data-driven foundation, guaranteeing that the project selected will maximize return on investment and be in line with strategic objectives **(APM, 2023)**.

Project evaluation forms prevent projects from being chosen based on bias or intuition. It compels a methodical assessment of how effectively the SOS app concept satisfies community needs in comparison to alternative concepts. By clearly contrasting the anticipated benefits—such as increased community safety and maybe saved lives—against the expenses, the form aids in the justification of the substantial resources committed. For the team to determine whether the project is worthwhile despite these obstacles, a Project Evaluation Form is required to highlight potential risks such as network reliability in rural areas and data privacy concerns **(APM, 2023)**.

## Question 1.2

### What is Cross-Platform Development?

Using a single codebase to produce an application that can operate on various mobile operating systems—mostly Android and iOS—is known as cross-platform development. Developers utilize frameworks like React Native, Flutter, or Xamarin to write the code once and publish it everywhere, preserving a near-native look, feel, and performance, as opposed to creating two distinct native apps (one in Swift for iOS and one in Java/Kotlin for Android) **(TechTarget, 2024)**.

## How will Cross-Platform Development Contribute to Meeting Your Project Timelines?

As a result, development time will be shortened. Compared to creating two native apps, the initial development effort is essentially cut in half because only one set of basic functionalities (SOS button, location sharing, and contact notification) need to be implemented. After deployment, only one codebase needs to be updated with any problem fixes or feature additions, and both platforms will then see the changes. By guaranteeing that iOS and Android customers receive upgrades at the same time, this expedites the post-launch support phase and saves a significant amount of time during the project's 12-month lifespan **(Warp Development, 2024)**.

### Three potential risks associated with cross-platform development

1. Cross-platform programs occasionally perform marginally worse than native apps when it comes to extremely sophisticated, graphics-intensive, or processor-heavy operations. Advanced capabilities like real-time augmented reality overlays may be difficult to implement, even while the basic SOS functionality is supported properly.
2. The reliability and durability of the selected cross-platform framework (such as Flutter) is critical to the project's success. The team must rely on the quality and availability of a third-party plugin, which might not be as reliable or well-maintained as native APIs, if a particular device feature is required.
3. Native developers have instant access to new features when Google or Apple release an OS update. The integration of the newest platform capabilities into the SOS app may be delayed since cross-platform developers must wait for the framework (such as React Native) to upgrade its bridge to accommodate those new features.

**(Warp Development, 2024)**

## Explain how Cross-Platform Development will Enhance the User Experience and Usability of your Application.

The user interface and experience will be the same for all users, regardless of whether they are using iOS or Android. In an emergency app where confusion must be kept to a minimum, this consistency is essential. Switching phone brands won't need a user to relearn how to send an SOS signal, making the experience more dependable and predictable **(Edwards, 2024)**.

The app instantly becomes available to almost every smartphone user in the community by launching on both main platforms at the same time on day one. This universal accessibility is essential to the app's main goal, which is to maximize its usability and public safety impact by making sure that no one is left without help because of their device's operating system **(Edwards, 2024)**.

## Question 1.3

### Project Purpose & Business Case

The project's existence is formally justified by this deliverable. It will highlight the community's main safety worry, which is the absence of a quick, centralized method for calling for help in an emergency. The business case will list the concrete and intangible advantages, like quicker emergency response times, heightened resident security, and the possibility of saving lives. It acts as the cornerstone document that outlines the reasons why the R1.75 million investment is both essential and beneficial for the community's well-being, directly relating to the main objective of the technology program **(Gillis, 2025)**.

### Project Scope & Key Deliverables

This part establishes the project's parameters, clearly outlining what will and won't be included to avoid "scope creep." The project will include the creation of a cross-platform mobile application called the "Guardian-SOS" app, which will have essential features such as an emergency contact notification system, GPS/SMS position sharing, and a one-tap SOS button. Importantly, to set clear expectations for stakeholders, it will also list exclusions, such as the production of any physical hardware devices, integration with official citywide emergency service dispatches, or the provision of on-the-ground medical assistance **(Gillis, 2025)**.

### Milestone Schedule

The 12-month project is broken down into important, quantifiable phases in this deliverable, which offers a timeline overview. It will involve significant achievements like "Completion of UX/UI Wireframes and Prototypes by End of Month 2," "Core App Development and Alpha Testing Completed by Month 6," "Public Beta Release and User Acceptance Testing (UAT) by Month 9," along with "Final App Launch on Google Play and Apple App Store by Month 12." This timetable guarantees that progress can be monitored monthly and offers distinct benchmarks for stakeholder evaluations **(Gillis, 2025)**.

## Budget & Resource Allocation

This gives the core team structure and financial plan approval. It will verify that the R1,750,000 fixed budget has been allotted to the following important areas: team salaries, software licenses (such as design tools and development frameworks), testing equipment, and a contingency fund. Additionally, it will legally approve the creation of the core project team, defining the four essential roles (e.g., Project Manager, Lead Cross-Platform Developer, UX/UI Designer, and QA Tester) and giving the project manager the power to appoint people to these roles **(Gillis, 2025)**.

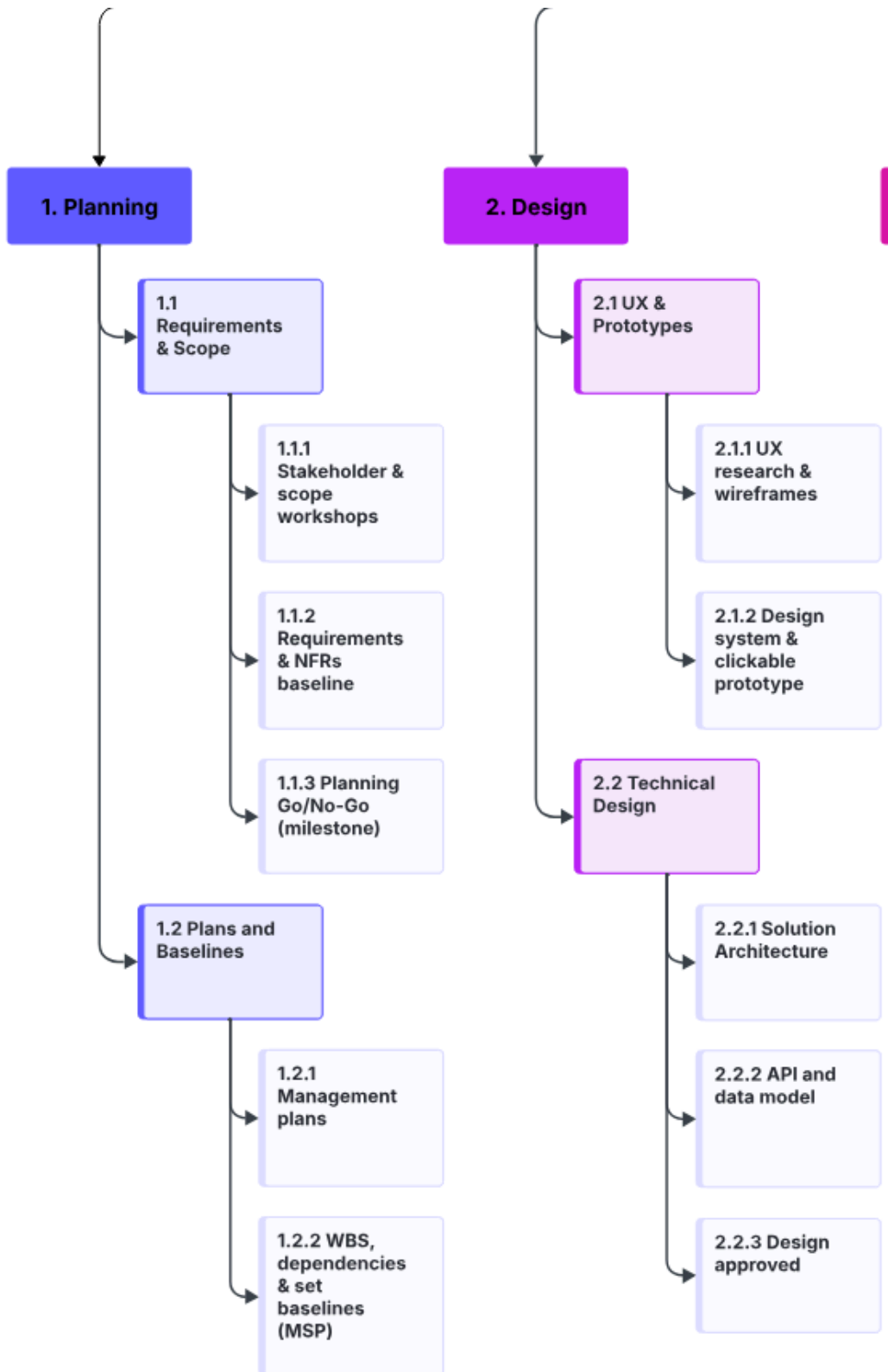
## Project Manager Appointment & Authority Level

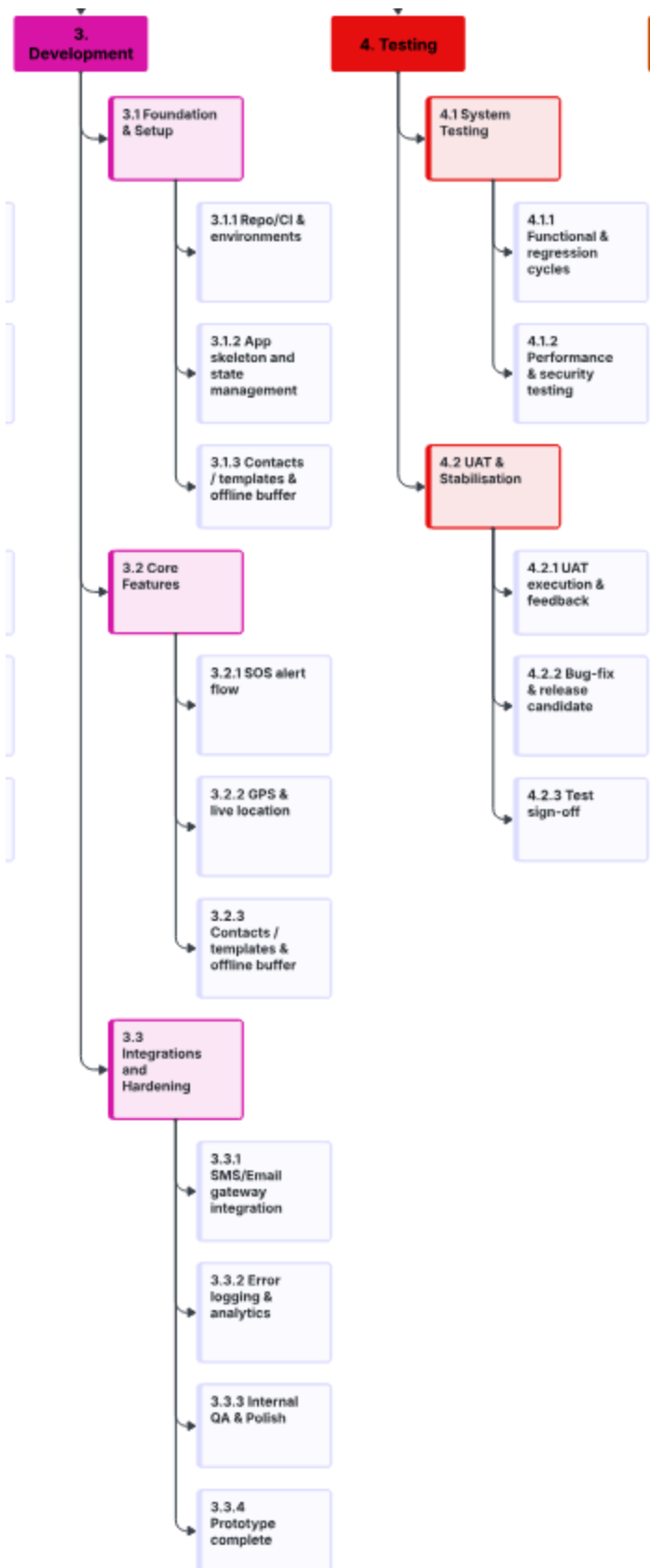
The Project Manager is officially named and given the express power to oversee the project in this crucial administrative deliverable. It will outline the boundaries of their decision-making authority, including their capacity to manage the project schedule, make daily team management decisions, and authorize expenditures up to a specific amount without executive approval. The project's steering committee or other important stakeholders must formally be notified of and approve any modifications that impact the project's scope, budget, or important dates **(Gillis, 2025)**.

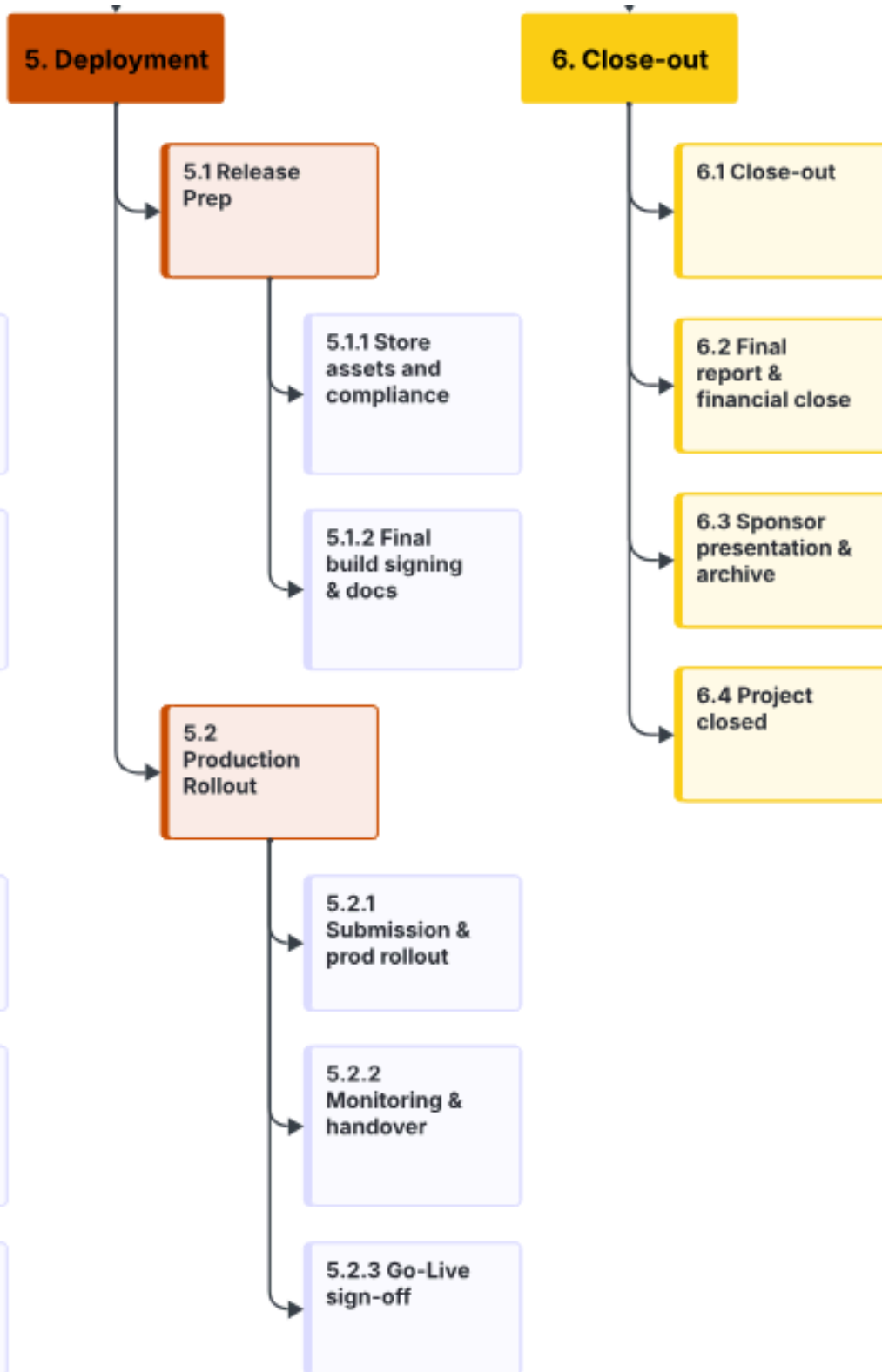
# Work Breakdown Structure Diagram



## Section Breakdown (For Ease of Marking)







# MS Project Solution

See *GitHub* for full project solution:

<https://github.com/ethan-edwards102/INSY6212-Assignment-2>

## Resource Assignments

| Name               | Role                    | Acronym |
|--------------------|-------------------------|---------|
| Dylan Olsen        | UI/UX Designer          | U       |
| Tyreece Sewbaran   | Software Developer      | S       |
| Ndumiso Jwara      | DevOps Engineer         | D       |
| Ethan Edwards      | Project Manager         | P       |
| David Foster       | QA Tester               | Q       |
| Justin Case (Fake) | Finance / Admin Officer | F       |

## Timeline (Split for Ease of Marking)

| WBS   | Name                                           | Cost    | Resource_Initials |
|-------|------------------------------------------------|---------|-------------------|
| 0     | Emergency SOS App                              | 1625920 |                   |
| 1     | Planning                                       | 132000  |                   |
| 1.1   | Requirements & Scope                           | 66000   |                   |
| 1.1.1 | Stakeholder & scope workshops                  | 30800   | P                 |
| 1.1.2 | Requirements & NFRs baseline                   | 35200   | P                 |
| 1.1.3 | Planning Go/No-Go (milestone)                  | 0       |                   |
| 1.2   | Plans & Baselines                              | 66000   |                   |
| 1.2.1 | Management plans<br>(schedule/cost/risk/comms) | 35200   | P                 |
| 1.2.2 | WBS, dependencies & set baselines (MSP)        | 30800   | P                 |
| 2     | Design                                         | 343200  |                   |
| 2.1   | UX & Prototypes                                | 190000  |                   |
| 2.1.1 | UX research & wireframes                       | 91200   | U,P               |
| 2.1.2 | Design system & clickable prototype            | 98800   | U,P               |
| 2.2   | Technical Design                               | 153200  |                   |

|       |                                          |                        |       |
|-------|------------------------------------------|------------------------|-------|
| 2.2.1 | Solution architecture (app/backend/auth) | 46080                  | S     |
| 2.2.2 | API & data model (+ NFR specs)           | 107120                 | S,P   |
| 2.2.3 | Design approved (milestone)              | 0                      |       |
| 3     | Development                              | 659200                 |       |
| 3.1   | Foundation & Setup                       | 105600                 |       |
| 3.1.1 | Repo/CI & environments                   | 33600                  | D     |
| 3.1.2 | App skeleton & state management          | 38400.0000<br>00000007 | S     |
| 3.1.3 | Security baseline & telemetry            | 33600                  | D     |
| 3.2   | Core Features                            | 343200                 |       |
| 3.2.1 | SOS alert flow                           | 114400                 | S,U,P |
| 3.2.2 | GPS & live location                      | 114400                 | P,S,U |
| 3.2.3 | Contacts/templates & offline buffer      | 114400                 | P,S,U |
| 3.3   | Integrations & Hardening                 | 210400                 |       |
| 3.3.1 | SMS/Email gateway integration            | 70400                  | S,U   |
| 3.3.2 | Error logging & analytics                | 67200                  | Q,S   |
| 3.3.3 | Internal QA & polish                     | 72800                  | P,Q   |

|       |                                    |        |         |
|-------|------------------------------------|--------|---------|
| 3.3.4 | Prototype complete (milestone)     | 0      |         |
| 4     | Testing                            | 268160 |         |
| 4.1   | System Testing                     | 159520 |         |
| 4.1.1 | Functional & regression cycles     | 88960  | P,Q,S   |
| 4.1.2 | Performance & security testing     | 70560  | D,Q,S   |
| 4.2   | UAT & Stabilization                | 108640 |         |
| 4.2.1 | UAT execution & feedback           | 58240  | P,Q     |
| 4.2.2 | Bug-fix sprint & release candidate | 50400  | D,S     |
| 4.2.3 | Test sign-off (milestone)          | 0      |         |
| 5     | Deployment                         | 189360 |         |
| 5.1   | Release Prep                       | 73360  |         |
| 5.1.1 | Store assets & compliance          | 44400  | D,U,P,S |
| 5.1.2 | Final build signing & docs         | 28960  | D,P,Q,S |
| 5.2   | Production Rollout                 | 116000 |         |
| 5.2.1 | Submission & prod rollout          | 58000  | D,P,S   |
| 5.2.2 | Monitoring & handover              | 58000  | P,D,S   |

|       |                                |       |     |
|-------|--------------------------------|-------|-----|
| 5.2.3 | Go-Live sign-off (milestone)   | 0     |     |
| 6     | Project Closure                | 34000 |     |
| 6.1   | Close-out                      | 0     |     |
| 6.2   | Final report & financial close | 20400 | F,P |
| 6.3   | Sponsor presentation & archive | 13600 | P,F |
| 6.4   | Project closed (milestone)     | 0     |     |

| WBS   | Name                          | Duration | Predecessors | Start_Date              | Finish_Date                |
|-------|-------------------------------|----------|--------------|-------------------------|----------------------------|
| 0     | Emergency SOS App             | 195 days |              | January 1, 2026 8:00 AM | September 30, 2026 5:00 PM |
| 1     | Planning                      | 30 days  |              | January 1, 2026 8:00 AM | February 11, 2026 5:00 PM  |
| 1.1   | Requirements & Scope          | 15 days  |              | January 1, 2026 8:00 AM | January 22, 2026 8:00 AM   |
| 1.1.1 | Stakeholder & scope workshops | 7 days   |              | January 1, 2026 8:00 AM | January 9, 2026 5:00 PM    |



|       |                                             |         |    |                           |                           |
|-------|---------------------------------------------|---------|----|---------------------------|---------------------------|
| 1.1.2 | Requirements & NFRs baseline                | 8 days  | 3  | January 12, 2026 8:00 AM  | January 21, 2026 5:00 PM  |
| 1.1.3 | Planning Go/No-Go (milestone)               | 0 days  | 4  | January 22, 2026 8:00 AM  | January 22, 2026 8:00 AM  |
| 1.2   | Plans & Baselines                           | 15 days |    | January 22, 2026 8:00 AM  | February 11, 2026 5:00 PM |
| 1.2.1 | Management plans (schedule/cost/risk/comms) | 8 days  | 5  | January 22, 2026 8:00 AM  | February 2, 2026 5:00 PM  |
| 1.2.2 | WBS, dependencies & set baselines (MSP)     | 7 days  | 7  | February 3, 2026 8:00 AM  | February 11, 2026 5:00 PM |
| 2     | Design                                      | 50 days | 1  | February 12, 2026 8:00 AM | April 23, 2026 8:00 AM    |
| 2.1   | UX & Prototypes                             | 25 days |    | February 12, 2026 8:00 AM | March 18, 2026 5:00 PM    |
| 2.1.1 | UX research & wireframes                    | 12 days | 8  | February 12, 2026 8:00 AM | February 27, 2026 5:00 PM |
| 2.1.2 | Design system & clickable prototype         | 13 days | 11 | March 2, 2026 8:00 AM     | March 18, 2026 5:00 PM    |

|       |                                          |         |    |                        |                        |
|-------|------------------------------------------|---------|----|------------------------|------------------------|
| 2.2   | Technical Design                         | 25 days |    | March 19, 2026 8:00 AM | April 23, 2026 8:00 AM |
| 2.2.1 | Solution architecture (app/backend/auth) | 12 days | 12 | March 19, 2026 8:00 AM | April 3, 2026 5:00 PM  |
| 2.2.2 | API & data model (+ NFR specs)           | 13 days | 14 | April 6, 2026 8:00 AM  | April 22, 2026 5:00 PM |
| 2.2.3 | Design approved (milestone)              | 0 days  | 15 | April 23, 2026 8:00 AM | April 23, 2026 8:00 AM |
| 3     | Development                              | 60 days | 9  | April 23, 2026 8:00 AM | July 15, 2026 5:00 PM  |
| 3.1   | Foundation & Setup                       | 20 days |    | April 23, 2026 8:00 AM | May 20, 2026 5:00 PM   |
| 3.1.1 | Repo/CI & environments                   | 10 days | 16 | April 23, 2026 8:00 AM | May 6, 2026 5:00 PM    |
| 3.1.2 | App skeleton & state management          | 10 days | 16 | April 23, 2026 8:00 AM | May 6, 2026 5:00 PM    |
| 3.1.3 | Security baseline & telemetry            | 10 days | 20 | May 7, 2026 8:00 AM    | May 20, 2026 5:00 PM   |
| 3.2   | Core Features                            | 20 days |    | May 21, 2026 8:00 AM   | June 17, 2026 5:00 PM  |

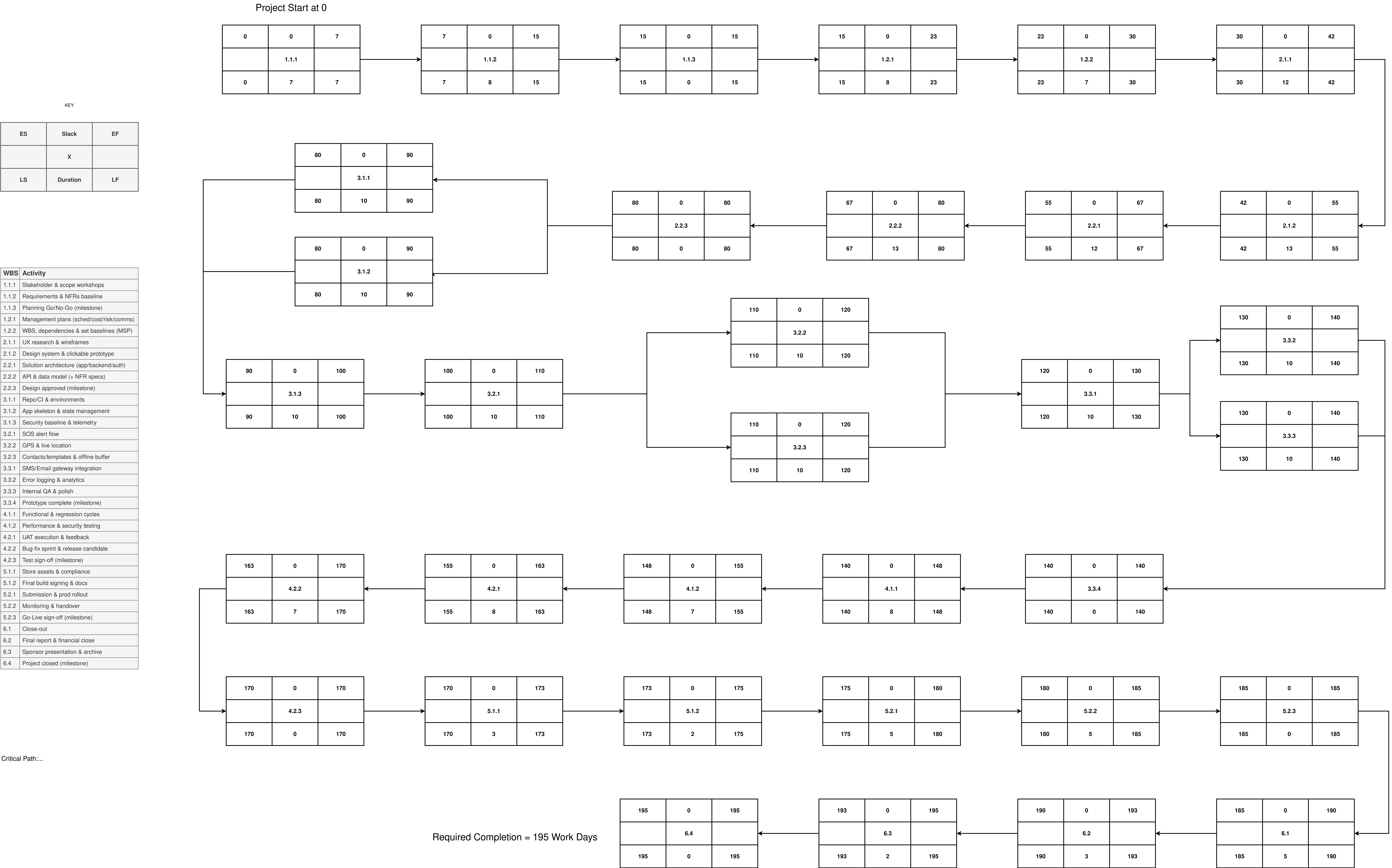
|       |                                     |         |    |                          |                            |
|-------|-------------------------------------|---------|----|--------------------------|----------------------------|
| 3.2.1 | SOS alert flow                      | 10 days | 21 | May 21, 2026<br>8:00 AM  | June 3, 2026<br>5:00 PM    |
| 3.2.2 | GPS & live location                 | 10 days | 23 | June 4, 2026<br>8:00 AM  | June 17, 2026<br>5:00 PM   |
| 3.2.3 | Contacts/templates & offline buffer | 10 days | 23 | June 4, 2026<br>8:00 AM  | June 17, 2026<br>5:00 PM   |
| 3.3   | Integrations & Hardening            | 20 days |    | June 18, 2026<br>8:00 AM | July 15, 2026<br>5:00 PM   |
| 3.3.1 | SMS/Email gateway integration       | 10 days | 25 | June 18, 2026<br>8:00 AM | July 1, 2026<br>5:00 PM    |
| 3.3.2 | Error logging & analytics           | 10 days | 27 | July 2, 2026<br>8:00 AM  | July 15, 2026<br>5:00 PM   |
| 3.3.3 | Internal QA & polish                | 10 days | 27 | July 2, 2026<br>8:00 AM  | July 15, 2026<br>5:00 PM   |
| 3.3.4 | Prototype complete (milestone)      | 0 days  | 29 | July 15, 2026<br>5:00 PM | July 15, 2026<br>5:00 PM   |
| 4     | Testing                             | 30 days | 17 | July 16, 2026<br>8:00 AM | August 26, 2026<br>5:00 PM |
| 4.1   | System Testing                      | 15 days |    | July 16, 2026<br>8:00 AM | August 5, 2026<br>5:00 PM  |

|       |                                    |         |    |                              |                               |
|-------|------------------------------------|---------|----|------------------------------|-------------------------------|
| 4.1.1 | Functional & regression cycles     | 8 days  | 30 | July 16, 2026<br>8:00 AM     | July 27, 2026<br>5:00 PM      |
| 4.1.2 | Performance & security testing     | 7 days  | 33 | July 28, 2026<br>8:00 AM     | August 5, 2026<br>5:00 PM     |
| 4.2   | UAT & Stabilization                | 15 days |    | August 6, 2026<br>8:00 AM    | August 26, 2026<br>5:00 PM    |
| 4.2.1 | UAT execution & feedback           | 8 days  | 34 | August 6, 2026<br>8:00 AM    | August 17, 2026<br>5:00 PM    |
| 4.2.2 | Bug-fix sprint & release candidate | 7 days  | 36 | August 18, 2026<br>8:00 AM   | August 26, 2026<br>5:00 PM    |
| 4.2.3 | Test sign-off (milestone)          | 0 days  | 37 | August 26, 2026<br>5:00 PM   | August 26, 2026<br>5:00 PM    |
| 5     | Deployment                         | 15 days | 31 | August 27, 2026<br>8:00 AM   | September 16, 2026<br>5:00 PM |
| 5.1   | Release Prep                       | 5 days  |    | August 27, 2026<br>8:00 AM   | September 2, 2026<br>5:00 PM  |
| 5.1.1 | Store assets & compliance          | 3 days  | 38 | August 27, 2026<br>8:00 AM   | August 31, 2026<br>5:00 PM    |
| 5.1.2 | Final build signing & docs         | 2 days  | 41 | September 1, 2026<br>8:00 AM | September 2, 2026<br>5:00 PM  |
| 5.2   | Production Rollout                 | 10 days |    | September 3, 2026<br>8:00 AM | September 16, 2026<br>5:00 PM |

|       |                                |         |    |                            |                            |
|-------|--------------------------------|---------|----|----------------------------|----------------------------|
| 5.2.1 | Submission & prod rollout      | 5 days  | 42 | September 3, 2026 8:00 AM  | September 9, 2026 5:00 PM  |
| 5.2.2 | Monitoring & handover          | 5 days  | 44 | September 10, 2026 8:00 AM | September 16, 2026 5:00 PM |
| 5.2.3 | Go-Live sign-off (milestone)   | 0 days  | 45 | September 16, 2026 5:00 PM | September 16, 2026 5:00 PM |
| 6     | Project Closure                | 10 days | 39 | September 17, 2026 8:00 AM | September 30, 2026 5:00 PM |
| 6.1   | Close-out                      | 5 days  | 46 | September 17, 2026 8:00 AM | September 23, 2026 5:00 PM |
| 6.2   | Final report & financial close | 3 days  | 48 | September 24, 2026 8:00 AM | September 28, 2026 5:00 PM |
| 6.3   | Sponsor presentation & archive | 2 days  | 49 | September 29, 2026 8:00 AM | September 30, 2026 5:00 PM |
| 6.4   | Project closed (milestone)     | 0 days  | 50 | September 30, 2026 5:00 PM | September 30, 2026 5:00 PM |

# Network Diagram





# GitHub Issues

Labels

Milestones

New issue

☐ Open 7

Closed 0

Author ▾ Labels ▾ Projects ▾ Milestones ▾ Assignees ▾ 

⌵ Newest ▾

☐

Project Closure

#7 · ST10247775 opened now

☐

Deployment

#6 · ST10247775 opened 1 minute ago

☐

Testing

#5 · ST10247775 opened 2 minutes ago

☐

Development

#4 · ST10247775 opened 3 minutes ago

☐

Design

#3 · ST10247775 opened 4 minutes ago

☐

Planning

#2 · ST10247775 opened 4 minutes ago

☐

Emergency SOS App

#1 · ST10247775 opened 1 hour ago



# References

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Warp Development. 2024. *Cross-Platform App Development: Advantages and Disadvantages*. 28 June 2024. [Online] Available at: <https://www.warpdevelopment.com/cross-platform-app-development-advantages-and-disadvantages/> (Accessed 06 October 2025).